B1 — 12 V DC POWER INPUT / PROTECTED HEATER SUPPLY

J2
12V DC input / + then 0V



F1
TBD current





D1
Reverse polarity / TBD rating



D2
TVS TBD

U1
12V –> 5V module / TBD



F1, D1, D2, U1 and Q1 require load current / transient / thermal sizing. D1 is a series diode, with conduction losses.
12 V prototype only. No 24 V qualification. Main load returns directly to input GND; keep analog ground paths quiet.

#FLG03
PWR_FLAG



B2 — PASSIVE MODULE INTERFACE / LOCAL 3.3 V

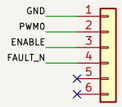
J1
A–B POWER / XH–2



J6
A–B I2C / XH–3

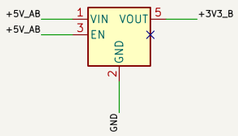


J7
A–B CTRL / XH–6





U2
AP2112K–3.3

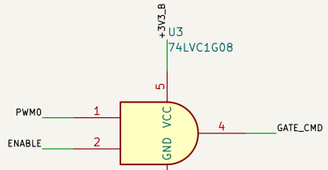




J1/J6/J7: same pin order as A. J7.5 / J7.6 reserved, not connected on B.
FAULT_N = ADC ALERT/RDY: configure comparator in firmware; no independent thermal cutoff.
All three cables: connect/disconnect ONLY with both boards unpowered. No 3V3 rail between A and B.

B3 — HEATER COMMAND / HARDWARE ENABLE

U3
74LVC1G08











Q1
NMOS / 3V3 gate / TBD



J3
Heater output

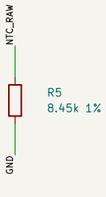


Gate drive = PWM0 AND ENABLE. Both default low with A unplugged.
A crashed with ENABLE high is not detected by this gate: add watchdog / independent thermal protection before unattended use.

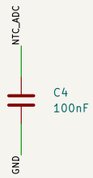
B4 — NTC INPUT / ADC / ADC ADDRESS 0x49

J4
NTC 15k B3920 probe

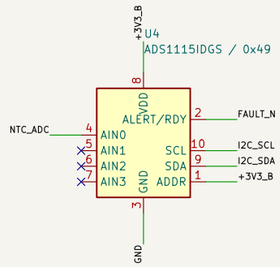


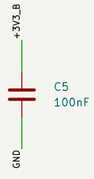


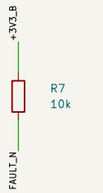




U4
ADS1115IDGS / 0x49







#FLG01
PWR_FLAG



#FLG02
PWR_FLAG



R_NTC = R5 * (Vexc / Vnode – 1). Calibrate R5 and Vexc.
Old 8400 ohm firmware constant must change to actual R5.
Probe fault / stale ADC: firmware must disable heater.
AIN1..3 unused. ADDR=VDD => 0x49. I2C pull–ups on A only.

DESIGN REVIEW ONLY — NOT FOR FABRICATION
Laboratory hardware platform

Sheet: /		
File: pcb_b_temperature.kicad_sch		
Title: PCB B — Temperature acquisition and heater power		
Size: A3	Date: 2026–11–04	**Rev: 0.2 DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/1